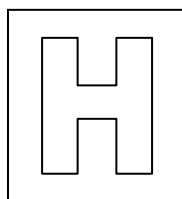


Candidate Name: _____

Class Adm No

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2011 Preliminary Examination II Pre-university 3

H2 Biology

9648/02

13 September 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This question paper consists of 24 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1.

- (a) Plant cells have a cellulose cell wall outside the cell surface membrane. The cellulose microfibrils are embedded in a matrix of hemicellulose, pectin and proteins.

- (i) Name the monosaccharide units that link together to form cellulose.

..... β-glucose; [1]

- (ii) Name the precise type of bond between the monosaccharide units in cellulose and describe how it is formed. Draw an annotated diagram in the space below to illustrate.

name of bond ... β (1 → 4) glycosidic bond;

formation ... condensation; (written and shown in illustration)
between C1 – C4; (written and shown in illustration)
correct labelling; (in illustration)
.....
.....
.....

[4]

- (b) The cell surface membrane of a plant cell contains cellulose synthase enzymes. These enzymes make cellulose microfibrils. The synthesised microfibrils pass out through the enzymes as they leave the cell. This is shown in Fig. 1.1.

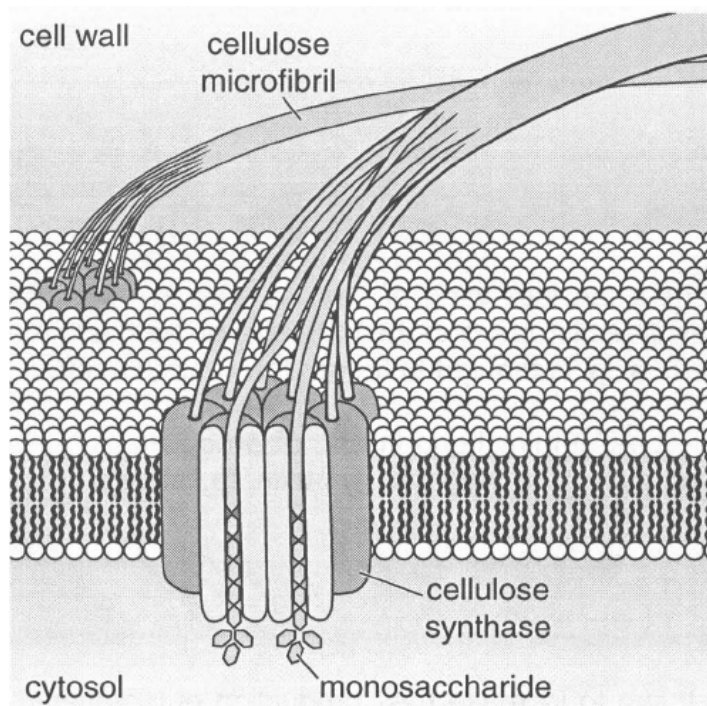


Fig. 1.1

- (i) Suggest why cellulose cannot be made outside the cell.

glucose monomers were in the cytosol, not outside the cell;
enzyme was situated in the cell surface membrane, with the active site facing the cytosol;

[1]

The cellulose fibres in the cell wall are made up of many cellulose microfibrils. Each microfibril is made up of many cellulose molecules lying parallel to each other.

- (ii) With reference to Fig. 1.1 and your knowledge of the structure of cellulose, explain how microfibrils are formed from cellulose molecules.

eight cellulose molecules leave the enzyme together;
Due to the alternate rotation of each glucose molecule, ;
hydrogen bonds are formed ;
between the OH groups of adjacent cellulose molecules;

[4]

- (c) The growth of plant cells is caused by a rapid increase in the size of the vacuole and requires the loosening of the cellulose fibres in the cell wall.

(i) Describe how water causes this increase in size of the vacuole to occur.

.....	osmosis, down a water potential gradient;	
...	entering the cell membrane;	
...		

..... [2]

- (ii) A fall in pH triggers the loosening of the cell wall by activating a protein called expansin. This causes disruption of the structure of the microfibrils. A plant growth substance, called auxin, is thought to trigger the change in pH.

From your knowledge of mitochondria and chloroplasts, suggest what process in the cell surface membrane is triggered by auxin, to cause a change in pH.

..	a change in pH would mean that the H ⁺ concentration needed to change.	
..	In chloroplasts or mitochondria this is brought about by	
	proton pumps; A chemiosmosis	[1]

[Total: 13]

2.

(a) Describe how DNA is packaged in eukaryotic cells.

- DNA molecules associated with histones to form nucleosome
- each nucleosome consist of DNA wrapped around eight histone molecules
- nucleosomes are linked to by a segment of DNA called linker DNA
- Histones have high proportion of positively charged amino acids (lysine and arginine), to bind tightly to the negatively charged DNA
- coils to form a fiber roughly 30nm in thickness, known as the 30-nm chromatin fiber / solenoid
- The 30-nm fiber, in turn, forms loops called looped domains, [which are attached to a chromosome scaffold made of non-histone proteins]
- This forms euchromatin which is less condensed.
- Some of the chromatin are more condensed forming heterochromatin.

[5]

(b) The ends of each DNA molecule are lengths of DNA called telomeres. In humans, telomeres are made up of the base sequence TTAGGG, which is repeated up to 400 times.

In some cells, each time a DNA molecule is replicated, the last 150 - 200 bases are not copied. The telomeres therefore get shorter with each DNA replication until they become too short for DNA replication to take place.

Calculate the **maximum** number of replications that can occur with the DNA that contains 400 TTAGGG repeats.

Show your working.

total number of bases involved was $400 \times 6 = 2400$. ;
If 2400 is divided by 150 (for the maximum number of replications) ;
the answer is 16. ;

[3]

- (c) In the body, only cells such as embryonic stem cells that need to divide regularly express telomerase. Other somatic cells do not normally express telomerase.

Most cancer cells which are continually dividing contain an enzyme, telomerase, which maintains the length of the telomere.

Suggest how these c

inhibitor of telomerase could be used

and explain how it might work

.....

.....

[2]

- (d) The enzyme that catalyses the replication of DNA checks for errors in the process and corrects them. This makes sure that the cells produced in mitosis are genetically identical.

Explain why checking for errors and correcting them is necessary.

accept ora
(errors are) mutations / named type of mutation ;
ora if corrected there are no mutations
(may lead to) production of altered proteins, so, impaired / loss, of function ;
A altered amino acid in, protein / primary structure
(may lead to) different antigens, so cells are rejected (by immune system) ;
idea that cells cannot function together / impaired coordination ;
ref. to cancerous cells / cancer(s) / tumours / sickle cell anaemia or other named monogenic condition ;
further detail ;
e.g. uncontrolled, division / mitosis / cell replication / cell growth
e.g. lack of contact inhibition / no apoptosis or described / (proto)oncogene(s)

[2]

[Total: 12]

3. Thyroid hormone, thyroxine, is produced by the thyroid gland, a gland in the throat region. Fig. 3.1 summarises the production and action of thyroxine in the body.

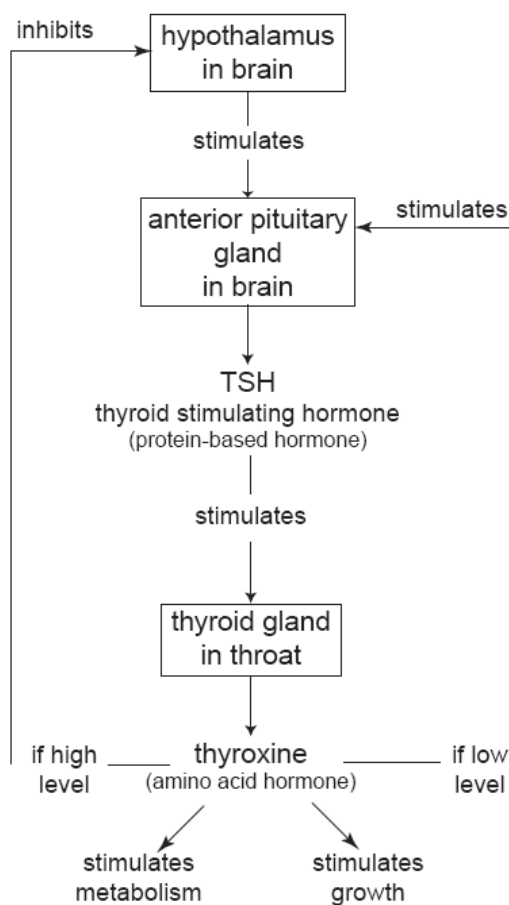


Fig. 3.1

- (a) Suggest how TSH travels to the thyroid gland.

Via the bloodstream;

[1]

The system shown in Fig. 3.1 is called a negative feedback system.

- (b) Describe a principle of a negative feedback system.

A system where the response to a particular stimulus reverses the direction of that stimulus;

[1]

Fig. 3.2 shows general models for signal reception and transduction

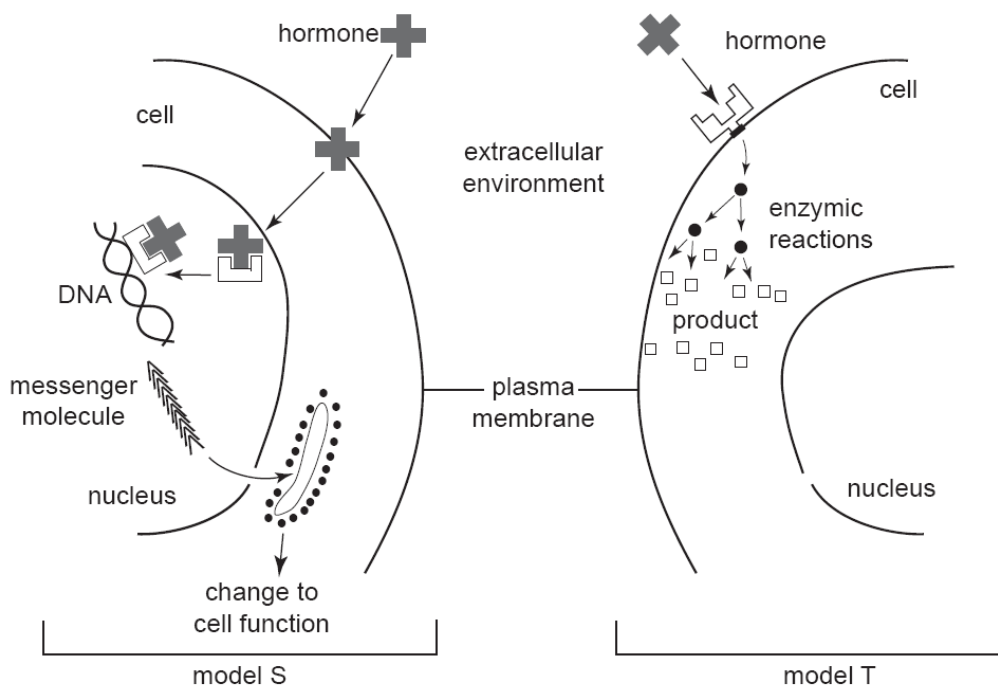


Fig. 3.2

(c) With reference to Fig. 3.2 and Fig. 3.1,

(i) which model, model S or model T, best corresponds to the mechanism of TSH?

..... Model T [1]

(ii) explain your answer in (i).

TSH is a protein-based hormone which is unable to pass directly through the plasma membrane due to size/hydrophilicity and as a consequence requires a membrane receptor.

To adequately answer this question students were required to provide information that was not already given in the stem of the question.

.....
.....
..... [1]

(d) Testes and brain tissue do not respond to thyroxine. Suggest why.

The cells in these organs lack the appropriate receptors for thyroxine.

..... [1]

The Fig. 3.3 shows some detail of a cell from a thyroid gland.

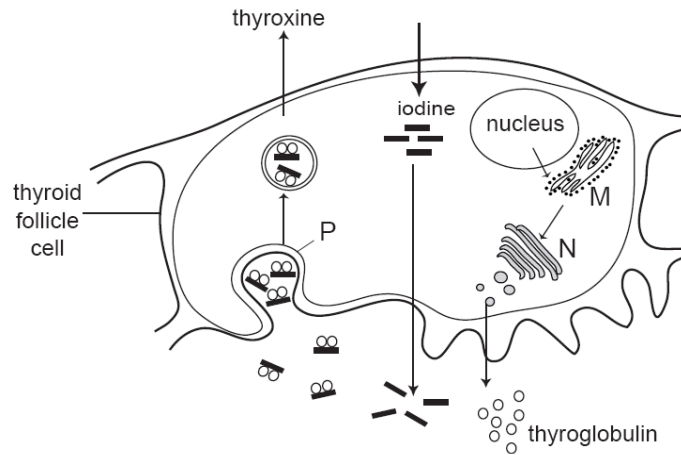


Fig. 3.3

(e) Outline the activity occurring in structure M.

Translation;
mRNA bound to fixed ribosome;
tRNA matching;
based on codon and anti-codon complementary base pairing;
polypeptide synthesised into cisternae;
transport vesicle pinches off, delivery to Golgi apparatus;

[4]

[Total: 9]

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*For
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4. Liver cells damaged by hepatitis infection switch on a gene called Fas, which causes them to selfdestruct. Pioneering research has produced a strikingly successful treatment for hepatitis in mice. The Fas gene was silenced by the technique of RNA interference.

RNA molecules, 21 to 23 nucleotides long, were injected into mice with hepatitis. The sequence of this 'small interfering RNA' (siRNA) matched part of the Fas gene. Once in the liver cell the two strands of the siRNA were separated so that one strand could bind to the mRNA transcript of the Fas gene. This caused the mRNA to be destroyed by enzymes, therefore preventing the gene product from being made.

This therapy prevented liver cell death and considerably increased the survival of mice with hepatitis.

(a)

- (i) Describe a way in which the function of mRNA differs from that of DNA.

.....	mRNA leaves nucleus ; <i>ora</i>	
.....	mRNA, translated / used to make, protein ;	
.....	DNA, transcribed / used to make, mRNA ;	
.....	mRNA short-term / DNA (long-term) store ;	.. [1]

- (ii) Describe two ways in which the structure of siRNA differs from mRNA.

1 ..	siRNA smaller / fewer nucleotides / only matches part of gene ; <i>ora</i>	
....	siRNA double-stranded ; <i>ora</i>	
....	Post-transcription modification;	
2 ..		
.....		 [2]

The technique of RNA interference has also been used to slow replication of HIV (Human Immunodeficiency Virus) *in vitro*. siRNA sequences that match the RNA genome of HIV can be used to trigger destruction of this RNA, preventing HIV from multiplying.

Another approach is to use RNA interference to silence genes for cell surface receptors, such as the CD4 and CCR5 molecules on human white blood cells. If these genes do not produce their protein antigens, HIV cannot bind to and infect the white blood cells.

- (b) Table 4.1 summarises some information about the two cell surface receptors used by HIV to bind to and infect white blood cells.

	cell surface receptor	
	CD4	CCR5
type of cell with this receptor	T lymphocyte white blood cells which divide by mitosis	macrophage cells which are long-lived and do not undergo mitosis
function of receptor	important roles in the immune system	limited, since 1% of people lack this receptor and show some resistance to HIV

Table 4.1

Experiments have been carried out where,

- siRNAs matching the CD4 mRNA were introduced into test tube populations of T lymphocytes;
- siRNAs matching the CCR5 mRNA were introduced into test tube populations of macrophages.

In both cases HIV was present but the presence of the siRNAs reduced its replication.

- (i) Use Table 4.1 to suggest with reasons which of the two test tube experiments showed most reduction of HIV replication.

.....	(CCR5 / macrophages)	
.....	(siRNAs continue to work) in long-lived cells ;	
.....	only one treatment needed for macrophages / CCR5 ;	
.....	(siRNAs diluted) as lymphocytes divide ; ora	
.....	repeat treatments needed for, lymphocytes / CD4 ;	

..... [2]

- (ii) Explain which receptor would be the best target for RNA interference if the approach was used as a therapy for humans infected with HIV.

.....	(CCR5)	
.....	because no essential function in body / absence not a problem ;	

..... [1]

- (c) Outline the process which occurs after HIV binds to its host.

...	Viral membrane fuse with host's membrane;	...
...	RNA and reverse transcriptase enter cell;	...
...	viral DNA made using viral RNA as template;	...
...	viral DNA incorporated into host DNA;	...
...	viral DNA used, for transcription/to form mRNA;	...
...	viral proteins made;	...
...	viral DNA used to produce RNA component of virus;	...
...	new viruses assembled;	...
...	viruses emerged by budding;	...

..... [5]

[Total:11]

5. Four different eye pigments in the fruit fly, *Drosophila melanogaster*, are made from the amino acid tryptophan. A simplified metabolic pathway of pigment production is shown in Fig. 5.1.

Three different gene loci control the pathway. Each locus has two alleles. These alleles are **V** or **v**, **C** or **c** and **B** or **b**, as shown in Fig. 5.1.

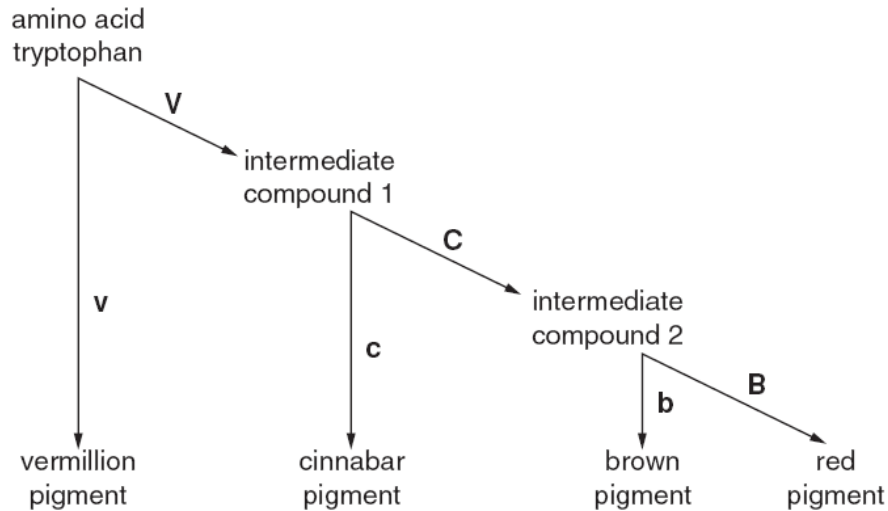


Fig. 5.1

(a)

- (i) Using the information in Fig. 5.1, deduce the phenotypes of flies with the following genotypes:

genotype	phenotype
VvCcBb	red ; vermillion ; cinnabar ;
vvCCBB	
VvccBB	

[3]

- (ii) State the term that is applied to this type of gene interaction.

..... recessive epistasis ; [1]

- (iii) Explain how the products **coded for** by the genes interact to give the different pigments.

- | | | |
|-------|--|-------|
| | 4. gene products are enzymes ; | |
| | 5. multi-enzyme / multi-step, pathway ; | |
| | 6. 3, steps / enzymes, change tryptophan to red pigment ; | |
| | 7. product of one reaction / intermediate compound, is, substrate / starting point, for next ; | |
| | 8. dominant allele gives, functional / wild-type / AW, enzyme ; | |
| | 9. recessive allele gives, non-functional / different / AW, enzyme ; | |

..... [3]

- (b) A mutation in another gene at another locus in *Drosophila* gives rise to white-eyed flies. The red eye allele of this gene (**R**) is known to be dominant to the white eye allele (**r**).

A student crossed a red-eyed fly with a white-eyed fly, expecting to get an F₁ generation of red-eyed flies. In fact, the results were as shown in Table 5.2.

phenotype of fly	number of offspring
red-eyed female	27
red-eyed male	0
white-eyed female	0
white-eyed male	23

Table 5.2

- (i) The student first suggested that the reason for there being red-eyed and white-eyed flies in the offspring was that the red-eyed parent was heterozygous.

Explain why this **cannot** be the correct explanation for the results shown in Table 5.2.

- if (red-eyed parent) was heterozygous
- 1. there would be no difference between, sexes / males and females ;
- 2. red-eyed males **and** white-eyed females would occur ; "because there are no red-eyed males and white-eyed females (in results)" "all 4 phenotypes would, occur / be represented"
- 3. 1:1:1:1 ratio
- **or**
- 1:1 ratio in both sexes ; .. [2]

- (ii) In *Drosophila*, the males are the heterogametic sex, possessing two different sex chromosomes, X and Y.

Draw a genetic diagram to show how the results shown in Table 5.2 could have been produced.

parental genotypes	XrXr	XRY-	;
gametes	Xr	XR and Y-	;
F1 genotypes	XRXr	XrY-	;
*Penalize for missing or wrong phenotypic ratio			

[3]

(iii) The chi-squared (χ^2) test can be used to analyse the results in Table 5.1.

The expected ratio of red-eyed females to white-eyed males is 1:1.

Use Table 5.2 to calculate a value for chi-squared (χ^2).

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad df = n - 1$$

Key to symbols:

Σ = 'sum of ...'

df = degrees of freedom

n = number of classes

O = observed value

E = expected value

Table 5.3

phenotype of fly	O	E	O - E	(O - E) ²	$\frac{(O - E)^2}{E}$
red-eyed female					
white-eyed male					

$$\chi^2 = \dots\dots\dots$$

Use your calculated value of χ^2 and the table of probabilities shown in Table 5.4 to test the significance of the difference between the observed and expected results.

degrees of freedom	probability, p			
	0.90	0.50	0.10	0.05
1	0.02	0.45	2.71	3.84
2	0.21	1.39	4.61	5.99

Table 5.4

State your conclusion below:

phenotype of fly	O	E	O - E	(O - E) ²	$\frac{(O - E)^2}{E}$
red-eyed female	27	25	2	4	0.16
white-eyed male	23	25	-2	4	0.16

$\chi^2 = 0.32$;

no significant difference (at 95% confidence level) ;

Differences due to chance;

Difference is statistically insignificant;

.....

.....

.....

.....

.....

..... [4]

[Total:16]

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Use*

6. An experiment was conducted to examine the effects of glucose and lactose on the levels of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals.

The results are shown in Fig. 6.1 below.

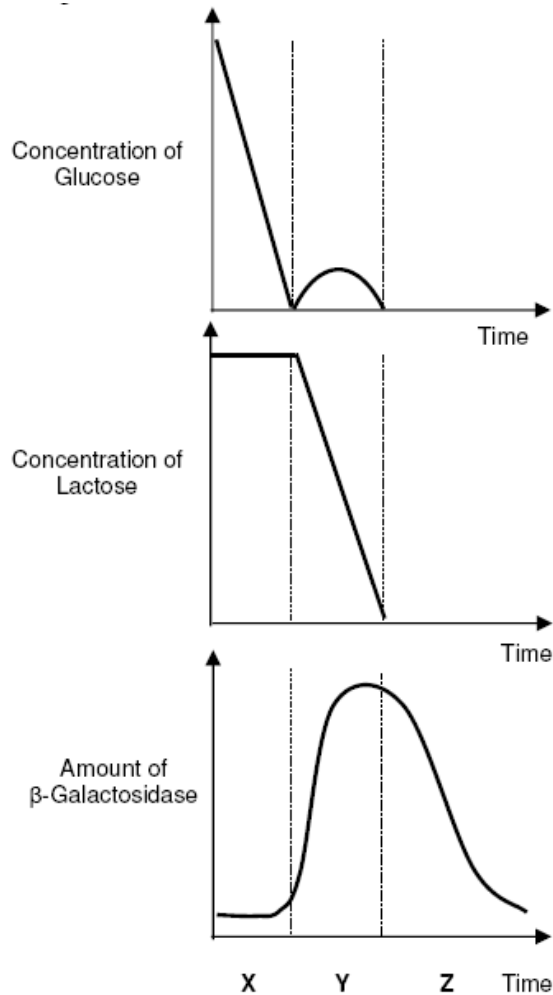


Fig. 6.1

- (a) State the two mechanisms of regulation of lac operon

...	Lac repressor binds to operator	...
...	CAP protein	...
...		...
...		...
.....		
.....		[2]

- (b) With reference to Fig. 6.1, explain how expression levels of β -galactosidase is regulated by the lac repressor in periods X and Y.

X (period X shows constant low level of β -galactosidase, but increase at the end of period X)
 High lactose level hence Lac repressor inactivated due to binding of allolactose;
 As glucose level becomes low, β -galactosidase is expressed;

Y (period Y shows constant increasing level of β -galactosidase, but peaked at the end of period Y)
 Lactose (including allolactose) digested, hence free up lac repressor;
 Activated Lac repressor bounds to operator, expression of β -galactosidase stopped;

 [4]

[Total:6]

7. The domestic dog, *Canis familiaris*, is one of the most striking examples of variation in a domestic animal because of the large number of different breeds. It was first domesticated at least 14 000 years ago and is thought to have descended from the grey wolf, *Canis lupus*.

- (a) Table 7.1 shows the classification of the grey wolf. Complete the table by writing your answers in the appropriate boxes.

kingdom		kingdom	Animalia
		phylum	Chordata
		class	Mammalia
		order	Carnivora
		family	Canidae
genus		genus	Canis
species		species	lupus / Canis lupus

1 mark for each correct column [2max]

Table 7.1

[2]

- (b) Define the term species.

.....	- capable of interbreeding ; A reproducing, mating - producing fertile offspring ; - belong to same gene pool / reproductively isolated ; - have a common ancestry ; - occupy the same (ecological) niche ; - have similar morphological, biochemical, behavioural, physiological and anatomical features (must name two of these for mark) ;	...
.....		...
.....		[2]

- (c) Five ancient breeds of dog are recognised, from which all other breeds are thought to have descended by artificial selection. One of these ancient breeds is the greyhound, *Canis familiaris leineri*. Most greyhounds are bred for racing.

Suggest how artificial selection in greyhounds has taken place.

- | | | |
|-------|--|-------|
| | <ul style="list-style-type: none"> • selected by humans ; • two examples of characteristic ; ; e.g. long back, muscular back, long legs, small feet, fast • allowed to, mate / AW ; | |
| | <ul style="list-style-type: none"> • offspring with ideal characteristics chosen to mate ; A best offspring chosen (repeated) over many generations ; | |
| | <ul style="list-style-type: none"> • allele frequency for desired characteristics increases / ref to directional selection ; • precautions against, inbreeding / selection of undesirable traits ; | |
| | | |

.....

.....

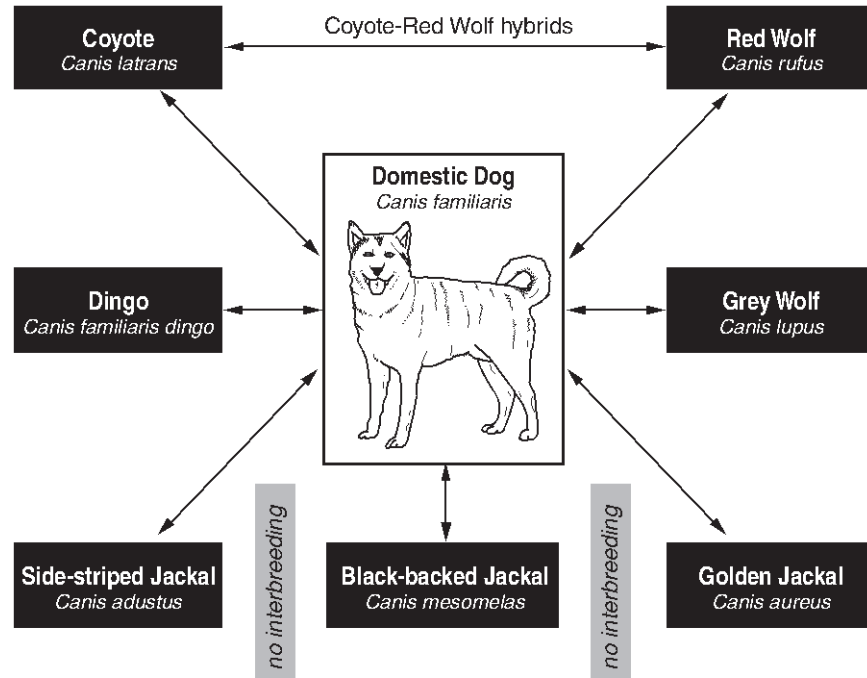
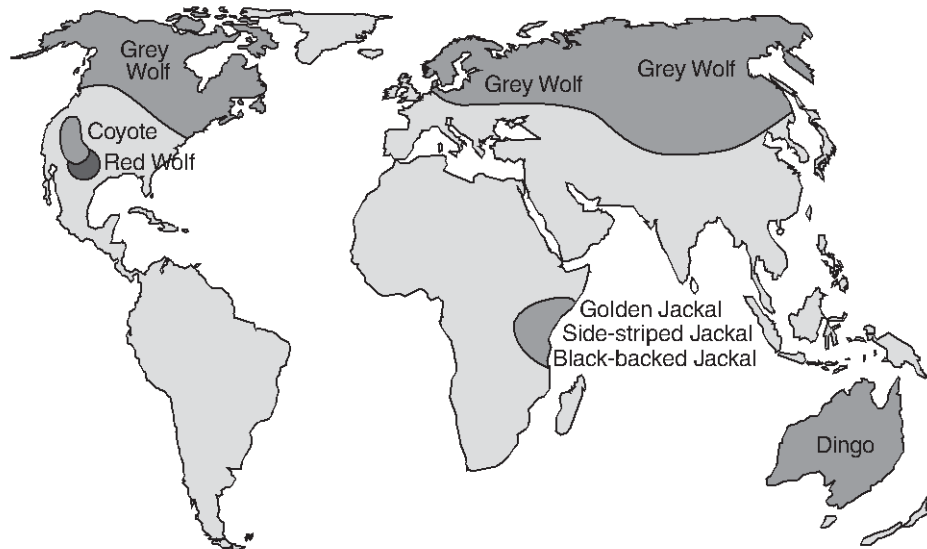
.....

..... [4]

- (d) The global distribution of most of the species belonging to the genus *Canis* is shown in Fig. 7.2.

As a result of the spread of human culture, distribution of the domestic dog is global. The domestic dog has been able to breed with all other members of the genus to form fertile hybrids.

The ranges of the three distinct species of jackal overlap in the Serengeti of East Africa.



Source: Advanced Biology A2 (2008). Page 189, BIOZONE International. By Tracey Greenwood, et al.

Fig. 7.2

- (i) State the type of isolating mechanism preventing, the three species of jackal from interbreeding

..... Jackal - sympatric / behavioural / ecological / temporal ; ignore reproductive isolation unqualified .. [1]

- (ii) State the type of isolating mechanism preventing, the dingo mating with all other

..... Dingo – geographical / allopatric ; [1]

- (iii) Using the information in Fig. 7.2, explain whether members of the genus *Canis* should be described as separate species.

- 1. all forms (fertile) hybrids with domestic dog ; A one named species forming (fertile)
 2. offspring with domestic dog (evidence suggests) domestic dog not separate species ; must be linked to mp1
 3. red wolf and coyote form hybrids ; R fertile hybrids
 4. may / may not, be fertile ;
 5. may / may not, be separate species ;
 6. (three types of) jackals do not interbreed ;
 7. (evidence suggests) they are separate species ; must be linked to 6
 [3]

[Total:13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8.

(a) Outline how the resting potential is maintained. [6]

(b) Describe the transmission of an action potential along a motor neurone. [8]

(c) Discuss the roles of synapses in the nervous system. [6]

[Total: 20]

9.

(a) Describe the main features of the Calvin Cycle. [8]

(b) Outline the role of NADP in photosynthesis. [6]

(c) Explain how ATP is synthesised using light energy in photosynthesis [6]

Qns 8a

[6 max]

1. reference to Na^+/K^+ pump;
2. active process/ATP used;
3. Na^+ (pumped) out and K^+ (pumped) in;
4. high Na^+ outside and high K^+ inside axon;
5. membrane slightly more leaky to K^+ /more K^+ leaks out than Na^+ leaks in/ reference to some K^+ channels open;
6. inside more negative than outside;

[Total: 20]

Qns 8b

(Motor neurone is myelinated)

[5 max]

1. Na^+ channels open ; **A** sodium channels
2. Na^+ enter cell ; **R** enter membrane
3. inside becomes, less negative / positive / +40mV / depolarised ;
4. Na^+ channels close ; **A** sodium channels
5. K^+ channels open ; **A** potassium channels
6. K^+ move out (of cell) ; **R** of membrane
7. inside becomes, negative / repolarised ; **A** negative figure

[3 max]

8. local circuits / description ;
9. (myelin sheath / Schwann cells) insulate axon / does not allow movement
10. of ions ;
11. action potential / depolarisation, only at nodes (of Ranvier) / gaps ;
12. saltatory conduction / **AW** ;
13. one-way transmission ;
14. **AVP** ; e.g. hyperpolarisation / refractory period

Qns 8c

(Question is about roles, not structure)

[6 max]

1. ensure one-way transmission ;
2. receptor (proteins) only in postsynaptic, membrane / neurone ; ora
3. vesicles only in presynaptic neurone ; ora
4. ref. adaptation ;
5. increased range of actions ;
6. due to interconnection of many nerve pathways ;
7. ref. inhibitory synapses ;
8. involved in memory / learning ;
9. due to new synapses being formed ;
10. **AVP**; e.g. summation / discrimination

BLANK PAGE**Qns 9a****[8 max]**

1. RuBP 5C ;
2. combines with carbon dioxide ;
3. rubisco ;
4. to form an unstable 6C compound ;
5. which forms 2 X GP (PGA) ;
6. ATP ;
7. and reduced NADP ;
8. forms TP (GAP) ;
9. TP used to form glucose / carbohydrates / lipids / amino acids ;
10. TP used in regeneration of RuBP
11. requires ATP ;
12. as source of phosphate ;
13. light independent ;

Qns 9b**[6 max]**

1. coenzyme ;
2. reduced ;
3. carries protons ;
4. and (high energy) electrons ;
5. from photosystem I / light stage ; R photosystem II
6. on thylakoid membrane grana ;
7. to stroma / Calvin cycle
8. ref. regeneration of NADP ;

Qns 9c**[6 max]**

1. light absorbed by chlorophyll / AW ;
2. ref. photosystems ;
3. ref. harvesting clusters / accessory pigments ;
4. reaction centre / P680 / P700 ;
5. excitation of electrons / AW ;
6. ETC ;
7. idea of different energy levels ;
8. $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$;
9. cyclic / non-cyclic, photophosphorylation ;
10. chemiosmosis / ATP synthase / description ;